



CSA0501-DATABASE MANAGEMENT SYSTEMS

LAB EXPERIMENTS

DATE: 4/08/2026

EXPERIMENT 15-- DCL COMMANDS-- GRANT, REVOKE

AIM:

To create and demonstrate an **AUTO_INCREMENT sequence** for generating register numbers in a database table.

PROCEDURE:

1. Create a MySQL user.
2. Grant privileges on the student_cgpa table to the user.
3. Verify the granted privileges.
4. Revoke the granted privilege from the user.
5. Verify that the privilege has been removed.

PROGRAM:

1. CREATE TABLE

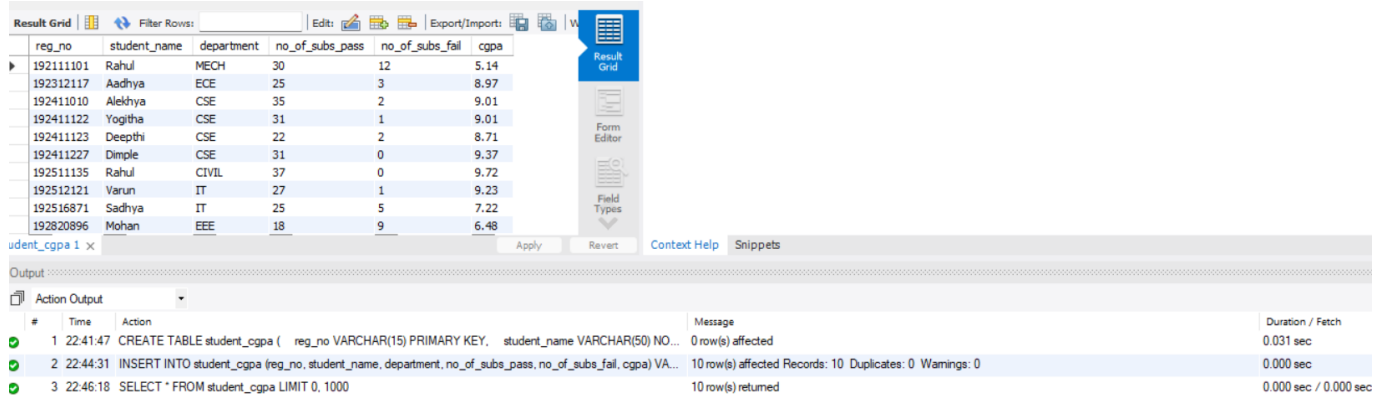
```
CREATE TABLE student_cgpa (  
    reg_no VARCHAR(15) PRIMARY KEY,  
    student_name VARCHAR(50) NOT NULL,  
    department VARCHAR(20),  
    no_of_subs_pass INT,  
    no_of_subs_fail INT,  
    cgpa DECIMAL(3,2)  
);
```

Then Insert Values:

```
INSERT INTO student_cgpa  
(reg_no, student_name, department, no_of_subs_pass, no_of_subs_fail, cgpa)  
VALUES  
(192411123,'Deepthi','CSE',22,2,8.71),  
(192512121,'Varun','IT',27,1,9.23),  
(192312117,'Aadhya','ECE',25,3,8.97),  
(192411227,'Dimple','CSE',31,0,9.37),  
(192820896,'Mohan','EEE',18,9,6.48),  
(192411122,'Yogitha','CSE',31,1,9.01),  
(192516871,'Sadhya','IT',25,5,7.22),  
(192111101,'Rahul','MECH',30,12,5.14),  
(192411010,'Alekhya','CSE',35,2,9.01),  
(192511135,'Rahul','CIVIL',37,0,9.72);
```

```
SELECT * FROM student_cgpa;
```

OUTPUT:



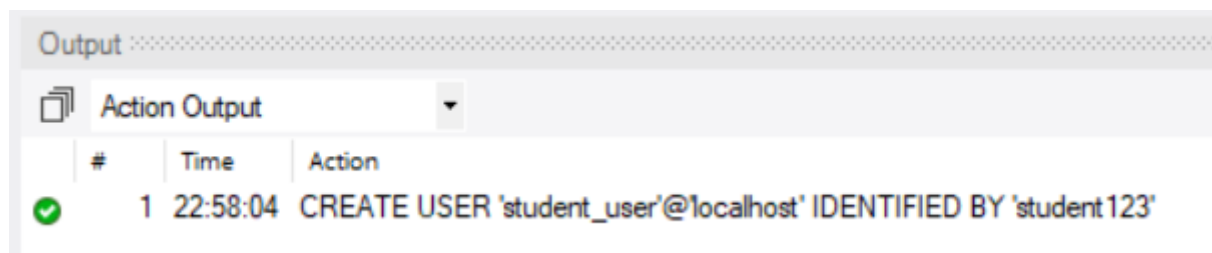
The screenshot shows a database management tool interface. At the top, there's a 'Result Grid' with columns: reg_no, student_name, department, no_of_subs_pass, no_of_subs_fail, and cgpa. Below the grid, there's an 'Output' section with a dropdown menu set to 'Action Output'. The output log shows three actions: 1. CREATE TABLE student_cgpa (reg_no VARCHAR(15) PRIMARY KEY, student_name VARCHAR(50) NO... 0 row(s) affected, 0.031 sec. 2. INSERT INTO student_cgpa (reg_no, student_name, department, no_of_subs_pass, no_of_subs_fail, cgpa) VA... 10 row(s) affected, Records: 10, Duplicates: 0, Warnings: 0, 0.000 sec. 3. SELECT * FROM student_cgpa LIMIT 0, 1000, 10 row(s) returned, 0.000 sec / 0.000 sec.

reg_no	student_name	department	no_of_subs_pass	no_of_subs_fail	cgpa
192111101	Rahul	MECH	30	12	5.14
192312117	Aadhya	ECE	25	3	8.97
192411010	Alekha	CSE	35	2	9.01
192411122	Yogitha	CSE	31	1	9.01
192411123	Deepthi	CSE	22	2	8.71
192411227	Dimple	CSE	31	0	9.37
192511135	Rahul	CIVIL	37	0	9.72
192512121	Varun	IT	27	1	9.23
192516871	Sadhya	IT	25	5	7.22
192820896	Mohan	EEE	18	9	6.48

2. Create User

```
CREATE USER 'student_user'@'localhost' IDENTIFIED BY 'student123';
```

OUTPUT:



The screenshot shows the 'Output' section of a database management tool. It displays a single action: 1. 22:58:04 CREATE USER 'student_user'@'localhost' IDENTIFIED BY 'student123'. The action is marked with a green checkmark, indicating success.

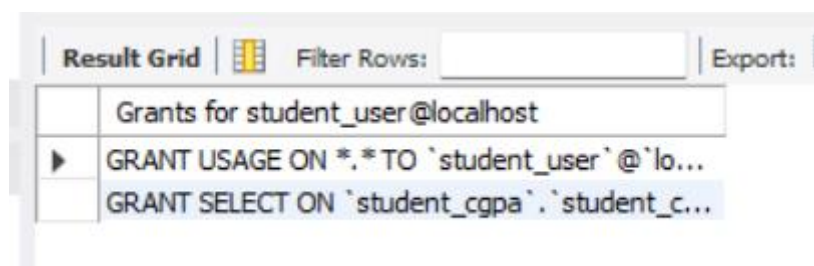
#	Time	Action
1	22:58:04	CREATE USER 'student_user'@'localhost' IDENTIFIED BY 'student123'

3. GRANT

Give the user permission to SELECT data from your student_cgpa table.

```
GRANT SELECT ON student_cgpa TO 'student_user'@'localhost';  
SHOW GRANTS FOR 'student_user'@'localhost';
```

OUTPUT:



The screenshot shows the 'Result Grid' of a database management tool. It displays the output of the SHOW GRANTS command for the user 'student_user'@'localhost'. The output shows two grants: GRANT USAGE ON *.* TO 'student_user'@'localhost' and GRANT SELECT ON 'student_cgpa' TO 'student_user'@'localhost'.

Grants for student_user@localhost
GRANT USAGE ON *.* TO 'student_user'@'localhost';
GRANT SELECT ON 'student_cgpa' TO 'student_user'@'localhost';

4. REVOKE

Take back the SELECT permission:

```
REVOKE SELECT ON student_cgpa FROM 'student_user'@'localhost';  
SHOW GRANTS FOR 'student_user'@'localhost';
```

OUTPUT:

Result Grid	Filter Rows:	Export: 
	Grants for student_user@localhost	
▶	GRANT USAGE ON *.* TO `student_user` @`lo...	

RESULT: Thus the DCL commands GRANT and REVOKE SQL executed successfully.